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Supplementary Materials for

Neural Decoding of Visual Imagery During Sleep

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Materials and Methods

The study protocol was approved by the Ethics Committee of ATR.

1. Subjects

Potential subjects answered questionnaires regarding their sleep-wake habits. Usual sleep and wake times, regularity of the sleep habits, habits of taking a nap, sleep complaints, and regularity of lifestyle (*e.g.*, mealtime), their physical and psychiatric health, and sleeping conditions were checked. Anyone who had physical or psychiatric diseases, was currently receiving medical treatment or was suspected of having a sleep disorder was excluded. People who had a habit of taking alcoholic beverages before sleep or smoking were also excluded. Finally, three healthy subjects (Japanese-speaking male adults, aged 27–39 years) with normal visual acuity participated in the experiments. All subjects gave written informed consent for their participation in the experiment.

2. Prior instructions to subjects

From three days prior to each experiment, subjects were instructed to maintain their sleep-wake habits, *i.e.*, daily wake/sleep time and sleep duration. They were also instructed to refrain from excessive alcohol consumption, unusual physical exercise, and taking of naps, from the day before each experiment. Their sleep-wake habits were monitored by a sleep log. No subject was chronically sleep deprived, and all slept over 6 hours on average the night before experiments.

3. Sleep adaptation

Subjects underwent two adaptation sleep experiments before the main fMRI sleep experiments to get used to sleeping in the experimental setting (*31, 32*). The adaptation experiments were conducted using the same procedures as the fMRI sleep experiments except that real fMRI scans were not performed. The experimental environment was simulated using a mock scanner consisting of the shell of a real scanner without the magnet. Echo-planar imaging (EPI) acoustic noise was also simulated and given to the subject via speakers.

4. Sleep experiment

Sleep (nap) experiments were carried out from 1:00 pm until 5:30 pm, and were scheduled to include the mid-afternoon dip (*33*). Subjects were instructed to sleep if they could, but not to try to sleep if they felt they could not. This instruction was given to reduce psychological pressure toward sleeping because efforts to sleep may themselves cause difficulty in falling asleep. fMRI scans were conducted simultaneous with PSG recordings (electroencephalogram [EEG], electrooculogram [EOG], electromyogram [EMG], and electrocardiogram [ECG]). We performed multiple awakenings (see below for details) to collect verbal reports on visual experience during sleep. The multiple-awakening procedure was repeated while fMRI was performed continuously (interrupted upon the subject's request for a break; mean ± SD of duration across all 55 runs, 88.99 ± 26.09 min [mean ± SD]). The experiment was repeated over 10, 7, and 7 days in Subject 1–3, respectively, until at least 200 awakenings with a visual report were obtained from each subject. Offline sleep stage scoring confirmed that in >90% of the awakenings followed by visual reports, the last 15-s epoch before awakening was classified as sleep stage 1 or 2 (fig. S2). If the last 15-s epoch was classified as the waking stage, we excluded the data from decoding analysis. As a result, 235, 198, and 186 awakenings were selected for Subject 1–3, respectively, constituting sleep data samples for further analysis.

Multiple-awakening procedure. Once the fMRI scan began, the subject was allowed to fall asleep. The experimenter monitored noise-reduced EEG recordings in real time while performing EEG staging on every 6-s epoch. The experimenter awakened subjects by calling them by name via a microphone/headphone when a single epoch with alpha-wave suppression and theta-wave (ripple) occurrence, which are known to be EEG signatures of NREM sleep stage 1 for obtaining frequent visual reports upon awakening (*11*), was detected (see fig. S1 for theta profiles). The subject was asked to verbally describe what they saw before awakening along with other mental content and then to go to sleep again. If the EEG signatures were detected before the elapsed time from the previous awakening reached 2 min, then the experimenter waited until it reached 2–3 min. If the subject was already in NREM sleep stage 2 when 2 min had passed, and it was unlikely they would go back to NREM sleep stage 1, the subject was awakened. When the subject repeatedly entered NREM sleep stage 2 within 2 min, the subject was awakened with short intervals (less than 2 min) or was asked to remain awake with eyes closed for one or two minutes after the awakening to increase their arousal level. This multiple-awakening procedure was repeated during the fMRI session. The subject was also asked to respond by button press when they detected a beep sound (250 Hz pure tone, 500 ms duration, 12–18-s inter-stimulus intervals). This auditory task was conducted for potential use in monitoring the sleep stage. However, in the present study, we did not use the data because we failed to record responses in some of the experiments owing to computer trouble. Our preliminary analysis using successfully recorded data (Subject 1) showed that the detection rates in each of the wake/sleep stages were similar to those of previous work (*34*) even when sleep samples were limited to those in which a sound was played during the last 15-s epoch before awakening but not detected by the subject, the decoding results were similar. Subjects were informed that they could quit the experiment anytime they wished to, and that they could refuse to report mental contents in cases where there were privacy concerns.

Acquisition of verbal reports. On each awakening, the subject was asked if the subject had seen anything just before awakening, and then to freely describe it along with other mental contents. If a description of the contents was unclear, the subject was asked to report the contents in more detail. Most reports started immediately upon awakening and lasted for 34 ± 19 s (mean ± SD; three subjects pooled). After the free verbal report, the subject was asked to answer specific questions such as rating the vividness of the image and the subjective timing of the experience (from when and until when relative to awakening), but the reports obtained by these explicit questions were not used in the analyses in the current study. Free reports that contained at least one visual element were classified as *visual reports*. If no visual content was present, reports were classified as *others* including *thought* (active thinking), *forgot*, *non-visual report*, and *no report*. The classification was first conducted in real time by the experimenter, and was later confirmed by other investigators. Examples of verbal reports are shown in table S1. The subject's voice during the procedure was recorded by an optical microphone.

5. MRI acquisition

fMRI data were collected using 3.0-Tesla scanner located at the ATR Brain Activity Imaging Center. An interleaved T2*-weighted gradient-EPI scan was performed to acquire functional images to cover whole brain (sleep experiments, visual stimulus experiments, and higher visual area localizer experiments: TR, 3,000 ms; TE, 30 ms; flip angle, 80 deg; FOV, 192 × 192 mm; voxel size, 3 × 3 × 3 mm; slice gap, 0 mm; number of slices, 50) or the entire occipital lobe (retinotopy experiments; TR, 2,000 ms; TE, 30 ms; flip angle, 80 deg; FOV, 192 × 192 mm; voxel size, 3 × 3 × 3 mm; slice gap, 0 mm; number of slices, 30). T2-weighted turbo spin echo images were scanned to acquire high-resolution anatomical images of the same slices used for the EPI (sleep experiments, visual stimulus experiments, and higher visual area localizer experiments: TR, 7,020 ms; TE, 69 ms; flip angle, 160 deg; FOV, 192 × 192 mm; voxel size, 0.75 × 0.75 × 3.0 mm; retinotopy experiments: TR, 6,000 ms; TE, 57 ms; flip angle, 160 deg; FOV, 192 × 192 mm; voxel size, 0.75 × 0.75 × 3.0 mm). T1-weighted magnetization-prepared rapid acquisition gradient-echo (MP-RAGE) fine-structural images of the whole head were also acquired (TR, 2,250 ms; TE, 3.06 ms; TI, 900 ms; flip angle, 9 deg, FOV, 256 × 256 mm; voxel size, 1.0 × 1.0 × 1.0 mm).

6. PSG recordings

PSG was performed simultaneously with fMRI. PSG consisted of EEG, EOG, EMG, and ECG recordings. EEGs were recorded at the 31 scalp sites in all experiments except for one (Fp1, Fp2, F7, F3, Fz, F4, F8, FC5, FC1, FC2, FC6, T7, C3, Cz, C4, T8, TP9, CP5, CP1, CP2, CP6, TP10, P7, P3, Pz, P4, P8, POz, O1, Oz, O2) according to 10% electrode positions (*35*). For one experiment, EEG was recorded at 25 scalp sites (Fp1, Fp2, F7, F3, Fz, F4, F8, T7, C3, Cz, C4, T8, P7, P3, Pz, P4, P8, PO7, PO3, POz, PO4, PO8, O1, Oz, O2). EOGs were recorded bipolarly from four electrodes placed at the outer canthi of both eyes (horizontal EOG) and above and below the right eye (vertical EOG). EMGs were recorded bipolarly from the mentum. ECGs were recorded from the lower shoulder blade. EEG and ECG recordings were referenced to FCz. Electrode impedance was kept below 20 kΩ. All data were recorded by an MRI-compatible amplifier at a sampling rate of 5,000 Hz using BrainVision Recorder. The artifacts derived from T2*-weighted gradient-EPI scan and ballistocardiogram (*36*) were reduced in real time using RecView so that the experimenter was able to monitor the EEG patterns online. Because EEG recordings were referenced to FCz, EEGs recorded in the occipital area were better than those recorded in the central area at detecting subtle changes in EEG waves. Thus, O1 was used for online monitoring of EEG data. When the O1 channel was contaminated by artifacts, O2 was used instead.

7. Offline EEG artifact removal and sleep-stage scoring

Artifacts were removed offline from EEG recordings after each experiment (the FMRIB plug-in for EEGLAB, The University of Oxford) for further analyses. EEG data were then down-sampled to 500 Hz, re-referenced to TP9 or TP10, depending on the derivation, and low-pass filtered with a 216-Hz cut-off frequency using a two-way least-squares FIR filter. EEG recordings were scored into sleep stages according to the standard criteria (*37*) for every 15 s.

8. Visual content labeling

The subject's report at each awakening, given verbally in Japanese, was transcribed into text. The reports that contained at least one visual object or scene were classified as *visual report*, and those without visual content were classified as *others*. Three labelers extracted words (nouns) that described visual objects or scenes from each visual report text and translated them into English (verified by a bilingual speaker). These words were mapped to *WordNet*, a lexical database in which words with a similar meaning are grouped as *synsets* in a hierarchical structure (*17*). Synset assignment was cross-checked by all three labelers. For each extracted word, hypernym synsets of the assigned synset were also identified in the *WordNet* hierarchy. To determine representative synsets that describe visual contents, we selected the synsets (synsets assigned to each word and their hypernyms) that were found in 10 or more visual reports without co-occurrence with at least one other synset. We then removed the synsets that were the hypernyms of others. These procedures produced *base synsets*, which were frequently reported while being semantically exclusive and specific. The visual contents at each awakening were coded by a *visual content vector*, in which the presence/absence of each synset was denoted by 1/0.

9. Visual stimulus experiment

We selected stimulus images for decoder construction using *ImageNet* (http://www.image-net.org/; 2011 fall release) (*19*), an image database in which web images are grouped according to *WordNet*. Two-hundred and forty images were collected for each base synset (each image corresponding to exclusively one synset). If images for a synset were not available in *ImageNet*, we collected images using *Google Images* (http://www.google.com/imghp). In this experiment, the selected images were resized so that the width and height of images were within 16 deg (the original aspect ratio was preserved) and were presented at the center of the screen on gray background. Subjects were allowed to freely view the images without fixation. We measured stimulus-induced fMRI activity for each base synset by presenting these images (using the same fMRI setting as the sleep experiment). In a 9-s stimulus block, six images were randomly sampled from the 192 (Subject 1) or 240-image (Subject 2 and Subject 3) set for one base synset without replacement, and each image was presented for 0.75 s with intervening blanks of 0.75 s. Thus each of the images for each base synset was presented only once during the experiment. The stimulus block (9 s) was followed by a 6-s rest period, and was repeated for all base synsets in each run. Extra 33-s and 6-s rest periods were added at the beginning and the end of each run, respectively. We repeated the run with different images to obtain 32, 40, and 40 blocks per base synset for Subject 1–3, respectively.

10. Localizer experiments

Retinotopy. The retinotopy mapping session followed the conventional procedure (*38, 39*) using a rotating wedge and an expanding ring of a flickering checkerboard. The data were used to delineate the borders between each visual cortical area, and to identify the retinotopic map on the flattened cortical surfaces of individual subjects.
Localizers for higher visual areas. We performed functional localizer experiments to identify the lateral occipital complex (LOC), fusiform face area (FFA), and parahippocampal place area (PPA) for each individual subject (*22, 23, 40*). The localizer experiment consisted of 4–8 runs and each run contained 16 stimulus blocks. In this experiment, intact or scrambled images (12 × 12 deg) of face, object, house and scene categories were presented at the center of the screen. Each of eight stimulus types (four categories × two conditions) was presented twice per run. Each stimulus block consisted of a 15-s intact or scrambled stimulus presentation. The intact and scrambled stimulus blocks were presented successively (the order of the intact and scrambled stimulus block was random), followed by a 15-s rest period of uniform gray background. Extra 33-s and 6-s rest periods were added at the beginning and end of each run, respectively. In each stimulus block, twenty different images of the same type were presented with a duration of 0.3 s followed by intervening blanks of 0.4-s duration. Images for each category were collected from the following resources: face images from THE CENTER FOR VITAL LONGEVITY (http://vitallongevity.utdallas.edu) (*41)*; object images from The Object Databank (http://stims.cnbc.cmu.edu/ImageDatabases/TarrLab/Objects/; Stimulus images courtesy of Michael J. Tarr, *Center*

for the Neural Basis of Cognition and Department of Psychology Carnegie Mellon University, http://www.tarrlab.org/); house and scene images from SUN database (http://groups.csail.mit.edu/vision/SUN/) (*42*).

11. MRI data preprocessing

The first 9-s scans for experiments with TR = 3 s (sleep, visual stimulus, and higher visual area localizer experiments) and 8-s scans for experiments with TR = 2 s (retinotopy experiments) of each run were discarded to avoid instability of the MRI scanner. The acquired fMRI data underwent three-dimensional motion correction by SPM5 (http://www.fil.ion.ucl.ac.uk/spm). The data were then coregistered to the within-session high-resolution anatomical image of the same slices used for EPI and subsequently to the whole-head high-resolution anatomical image. The coregistered data were then reinterpolated by 3 × 3 × 3 mm voxels. For the sleep data, after a linear trend was removed within each run, voxel amplitude around awakening was normalized relative to the mean amplitude during the period 60–90 s prior to each awakening. The average proportions of wake, stage 1, and stage 2 during this period were 32.7%, 44.0%, and 23.3%, respectively (three subjects pooled). This period was used as the baseline because it tended to show relatively stable BOLD signals over time. We assumed that the occurrence of sleep-onset imagery would be rare with the relatively low theta amplitudes (fig. S1) (*11*). However, it cannot be ruled out that visual imagery may have been experienced during this period, and that the baseline using this period as the baseline would make it difficult to detect visual contents relevant to the imagery in this period. The voxel values averaged across the three volumes (9 s) immediately before awakening served as a data sample for decoding analysis (the time window was shifted for time course analysis). For the visual stimulus data, after within-run linear trend removal, voxel amplitudes were normalized relative to the mean amplitude of the pre-rest period of each run and then averaged within each 9-s stimulus block (three volumes) shifted by 3 s (one volume) to compensate for hemodynamic delays. Voxels used for the decoding of a synset pair in each ROI were selected by *t* statistics comparing the mean responses to the images of paired synsets (highest absolute *t* values; 400 voxels for individual areas, and 1,000 voxels for LVC and HVC). The voxel values in each data sample were z-transformed for removing potential mean-level differences between the sleep and visual stimulus experiments.

12. Region of interest (ROI) selection

V1, V2, and V3 were delineated in the standard retinotopy experiment (*38, 39*), and the lateral occipital complex (LOC), the fusiform face area (FFA), and the parahippocampal place area (PPA) were identified using conventional functional localizers (*22*, *23*, *40*). The retinotopy experiment data were transformed to the Talairach coordinates and the visual cortical borders were delineated on the flattened cortical surfaces using BrainVoyager QX (http://www.brainvoyager.com). The voxel coordinates around the gray-white matter boundary in V1–V3 were identified and transformed back into the original coordinates of the EPI images. The voxels from V1, V2, and V3 were combined as the *lower visual cortex* (LVC; ~2,000 voxels in each subject). The localizer experiment data for higher visual areas were analyzed using SPM5. The voxels that showed significantly higher activation in response to objects, faces, or scenes than scrambled images for each (*t* test, uncorrected $P < 0.05$ or 0.01) were identified, and were used as ROIs for LOC, FFA, and PPA, respectively. We set relatively low thresholds for identifying ROIs so that a larger number of voxels would be included to preserve broadly distributed patterns. A continuous region covering LOC, FFA, and PPA was manually delineated on the flattened cortical surfaces, and the region was defined as the *higher visual cortex* (HVC; ~2,000 voxels in each subject). Voxels overlapping with LVC were excluded from HVC. After voxels of extremely low signal amplitudes were removed, approximately 2,000 voxels remained in LVC (2054, 2172, and 1935 voxels for Subject 1–3, respectively) and in HVC (1956, 1788, and 2235 voxels for Subject 1–3, respectively). For the analysis of individual subareas, the following numbers of voxels were identified for V1, V2, V3, LOC, FFA, and PPA, respectively: 885, 901, 728, 523, 537, and 353 voxels for Subject 1; 779, 949, 897, 329, 382, and 334 voxels for Subject 2; 710, 859, 765, 800, 432, and 316 voxels for Subject 3. LOC, FFA, and PPA identified by the localizer experiments may overlap: the voxels in the overlapping region were included in both ROIs.

13. Decoding analysis

For all pairs of the base synsets, a binary decoder consisting of linear Support Vector Machine (*20*) (SVM; implemented by LIBSVM (*43*)) was trained on the visual stimulus data of each ROI. The fMRI signals of the selected voxels and the synset labels were given to the decoder as training data. SVM provided a linear discriminant function for classification between synset k and l given input voxel values $\mathbf{x} = [x_1, \ldots, x_D]$ (D, number of voxels),

$$f_{kl}(\mathbf{x}) = \sum_{d=1}^{D} w_d x_d + w_0$$

where w_d is the weight parameter for voxel *d*, and w_0 is the bias. The performance was evaluated by the correct classification rate for all sleep samples selected for each pair. The synsets of a pair can have unbalanced numbers of samples, which could potentially lead to some bias. However, when the correct rates were calculated in each synset and then averaged, the averaged correct rates were highly correlated with the correct rates for all pooled samples (correlation coefficients for Subject 1–3, 0.96, 0.97, and 0.97, respectively). Therefore the bias, if any, is likely to be small. In the pairwise decoding analysis, the stimulus-trained decoder was tested on the sleep data that contained exclusively one of the paired synsets. Prediction was made on the basis of whether $f_{kl}(\mathbf{x})$ was positive (k) or negative (l), given a sleep fMRI data sample. In the multilabel decoding, the discriminant functions comparing a base synset k and each of the other synsets ($l \neq k$) were averaged after the normalization by the norm of the weight vector $\mathbf{w}_{kl}$ to yield the linear detector function (*5*), which indicates how likely synset k is to be present,

$$f_k(\mathbf{x}) = \frac{1}{N-1} \sum_{l \neq k} \frac{f_{kl}(\mathbf{x})}{\| \mathbf{w}_{kl} \|}$$

where N is the number of base synsets. Given a sleep fMRI data sample, multilabel decoding produced a vector consisting of the output scores of the detector functions for all base synsets [$f_1(\mathbf{x})$, $f_2(\mathbf{x})$, …, $f_N(\mathbf{x})$].

14. Synset pair selection by within-dataset cross-validation

To select synset pairs with content-specific patterns in both the stimulus-induced and sleep datasets, we performed cross-validation decoding analysis for each pair in each dataset. For the stimulus-induced dataset, samples from one run were left for testing, and the rest were used for decoder training (repeated until all runs were tested; leave-one-*run*-out cross validation) (*5*). For the sleep dataset, one sample was left for testing, and the rest were used for decoder training (leave-one-sample-out cross-validation). Note that since the frequency in the sleep reports is generally different between paired synsets, their available samples are usually unbalanced. To avoid possible biases in decoder training caused by the imbalance, we trained multiple decoders by randomly resampling a subset of the training data for the synset with more samples to match to the synset with fewer samples (repeated 11 times). The discriminant functions calculated for all the resampled training datasets were averaged after normalization by the norm of the weight vector to yield the discriminant function (decoder) to be used for testing in each cross-validation step. We selected the synset pairs that showed high cross-validation performance in both of datasets (one-tailed binomial test, uncorrected $P < 0.05$).

Supplementary Text for Online Movies

The movies illustrate the evolution of contents predicted by multilabel decoding for two individual sleep samples. The image panel displays stimulus images averaged with the weights according to the scores of the synsets at each time point. Stimulus images were randomly picked from the image samples for each synset. The graph shows the time courses of individual synset scores (red, reported synsets; black, unreported synsets). The tag clouds at the sides illustrate the names of the base synsets whose sizes vary according to the synset scores. The verbal reports for these sleep samples are as follows (note that the reports were originally Japanese):

> **Movie S1**. *"What I was just looking at was some kind of characters. There was something like a writing paper for composing an essay, and I was looking at the characters from the essay or whatever it was. It was in black and white and the writing paper was the only thing that was there. And shortly before that I think I saw a movie with a person in it or something but I can't really remember."*
>
> **Movie S2**. *"Well, there were persons, about 3 persons, inside some sort of hall. There was a male, a female, and maybe like a child. Ah, it was like a boy, a girl, and a mother. I don't think that there was any color."*

The report for Movie S1 contains three words describing visual objects or scene (*character*, *writing paper*, and *person*), and only *character* was assigned with a base synset (character synset [ID:06818970], *character*). The report for Movie S2 contains eight visual words (*person*, *hall*, *male*, *female*, *child*, *boy*, *girl*, and *mother*), and five out of the eight words were assigned with base synsets (male synset [ID:09624168], *male*, *boy*; female synset [ID:09619168], *female*, *girl*, *mother*). Note that several visual words were not assigned with base synsets, and thus, were not reflected in the movies (see footnote of table S1).

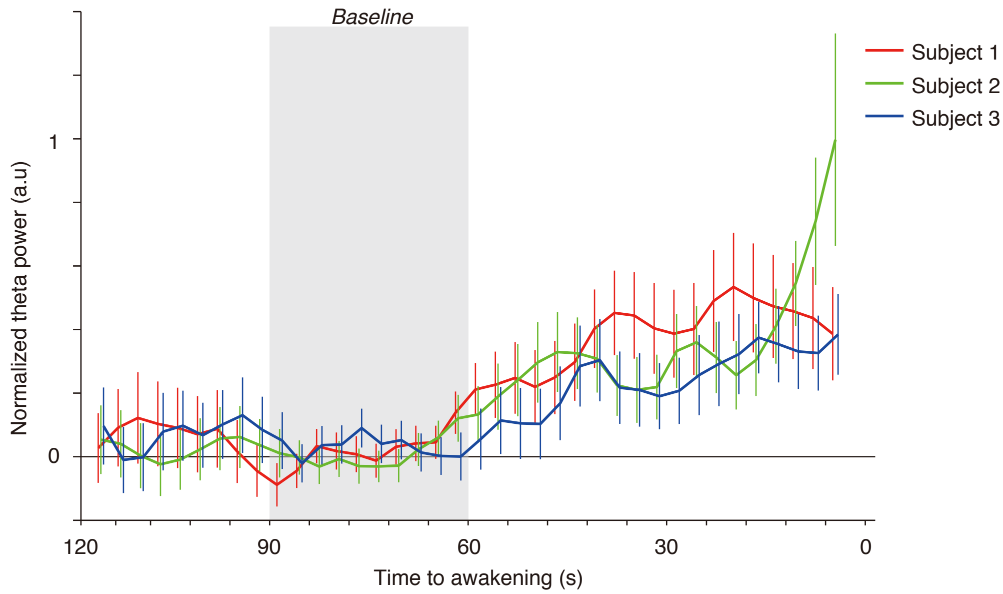

Fig. S1. Time course of theta power. The time course of theta power (4–7 Hz) during 2 min before awakening is shown for each subject (error bar, 95% CI; averaged across awakenings). For each awakening, we shifted the 9-s window (equivalent to three fMRI volumes) by 3 s (equivalent to one fMRI volume) to calculate the theta power from preprocessed EEG signals. The plotted points indicate the center of the 9-s window (slightly displaced to avoid overlaps between subjects). The power was normalized relative to the mean power during the time window of 60–90 s prior to each awakening (gray area). The result of this offline analysis is consistent with the awakening procedure in which theta ripples were detected in online monitoring to determine the awakening timing.

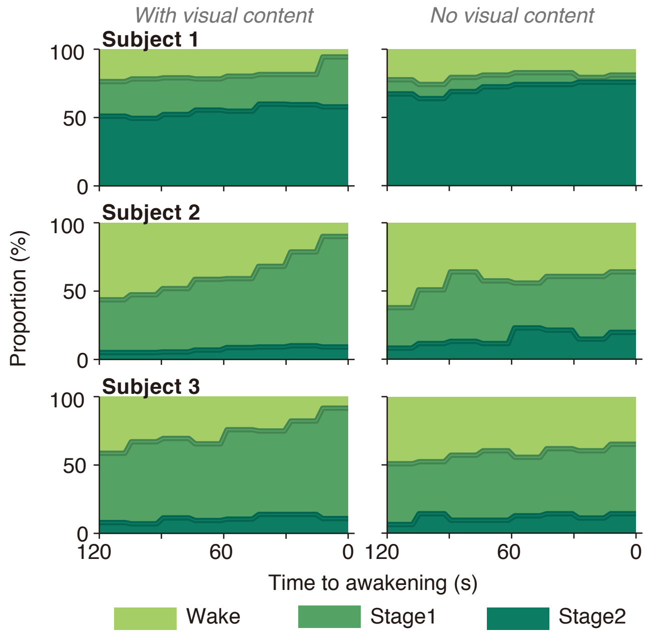

Fig. S2. Time course of sleep state proportion. The proportion of wake/sleep states (Wake, Stage 1, and Stage 2) is shown for all the awakenings with and without visual content in each subject. Offline sleep stage scoring was conducted for every 15-s epoch using simultaneously recorded EEG data. The last 15-s epoch before awakening was classified as Sleep Stage 1 or 2 in over 90% of the awakenings with visual contents, but in fewer awakenings with no visual content. This result suggests that most visual reports are indeed associated with imagery during sleep, not with imagery during wakefulness. The samples in which the last 15-s epoch before awakening was classified as Wake were not used for further analyses.

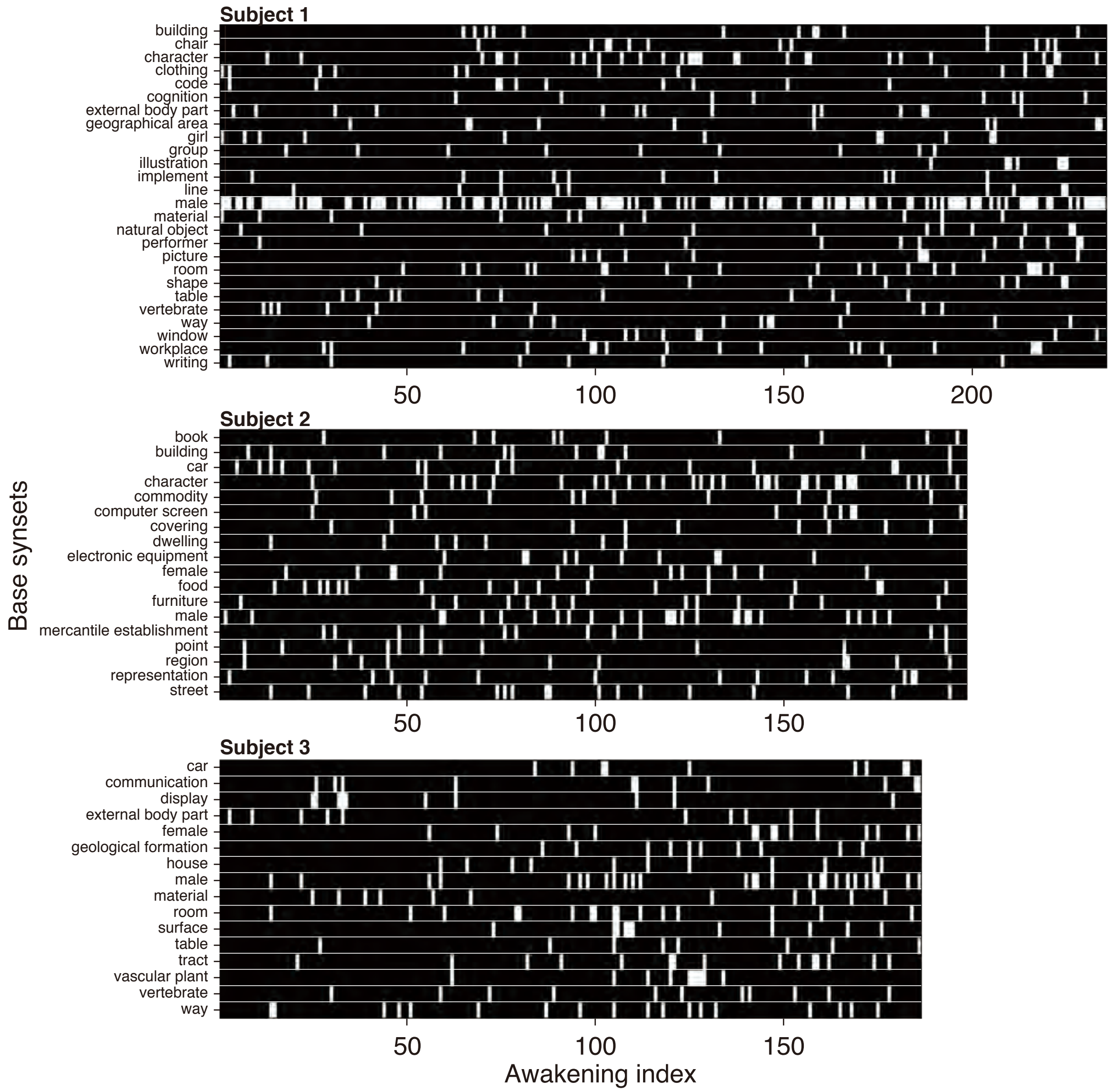

Fig. S3. Visual content vectors. Visual content vectors are shown for all subjects and awakenings with visual reports (excluding those with contamination of the wake stage detected by offline sleep staging). Each column denotes the presence/absence of the base synsets in the sleep sample. Note that there are several samples in which no synset is present. This is because the reported words in these samples were rare and not included in the base synsets.

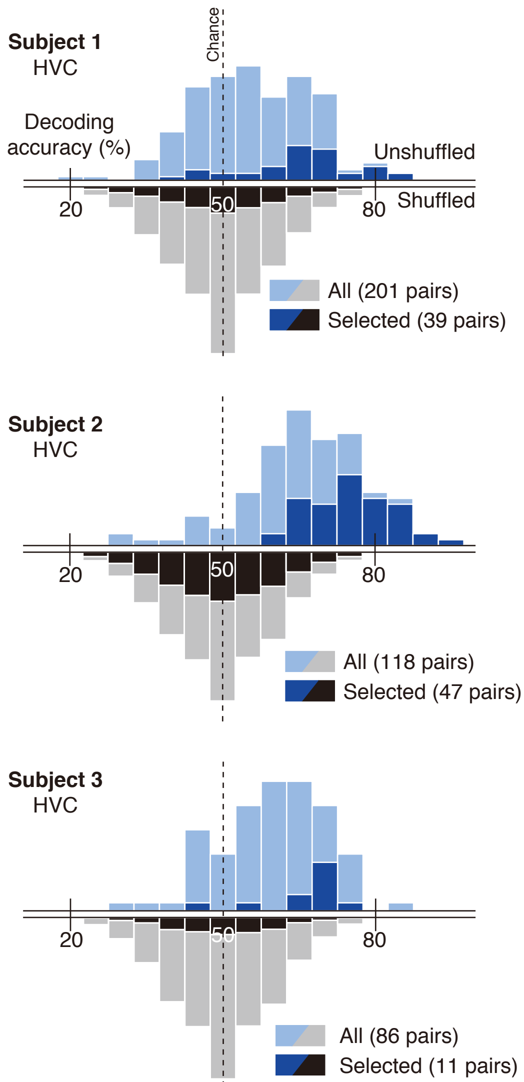

Fig. S4. Distribution of pairwise decoding accuracies. The distribution of the stimulus-to-sleep pairwise decoding accuracies for the higher visual cortex (HVC) is shown together with that from label-shuffled decoders. The format is the same as in Fig. 3B. The mean decoding accuracies for all pairs were 56.0% (95% CI [54.7, 57.3]) for Subject 1, 66.9% (CI [65.1, 66.8]) for Subject 2, and 59.8% (CI [57.9, 61.7]) for Subject 3, and those for selected pairs were 65.3% (CI [62.4, 68.2]) for Subject 1, 75.4% (CI [73.3, 77.4]) for Subject 2, and 66.1% (CI [61.2, 71.0]) for Subject 3. The fraction of pairs that showed significantly better decoding performance than chance level (one-tailed binomial test, uncorrected $P < 0.05$) was 22.4% (45/201) for Subject 1, 57.6% (68/118) for Subject 2, and 27.9% (24/86) for Subject 3. The performance was significantly better than that from label-shuffled decoders for all subjects for both of all and selected pairs (Wilcoxon rank-sum test, $P < 0.001$).

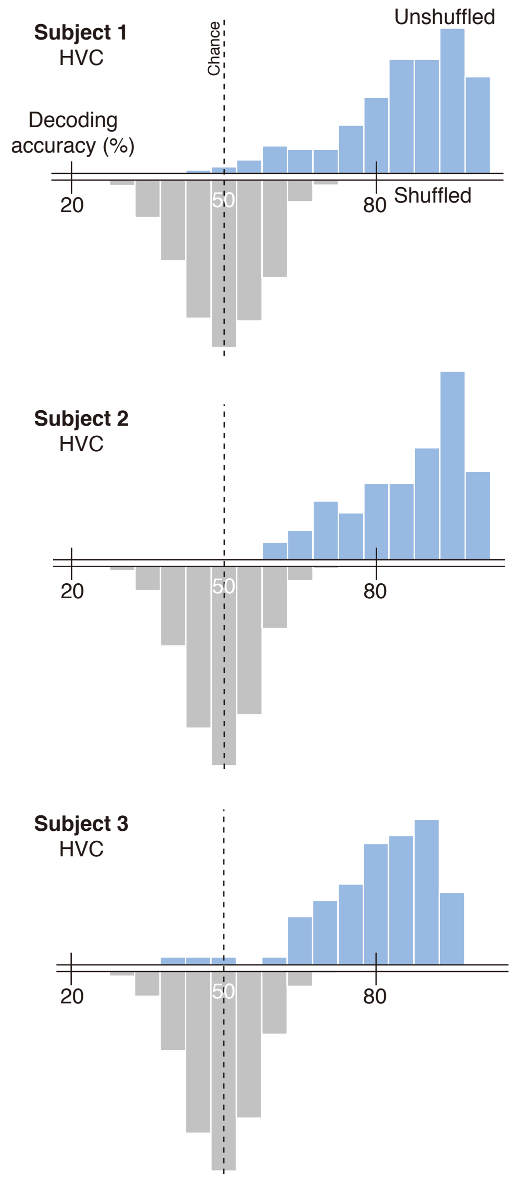

Fig. S5. Stimulus-to-stimulus pairwise decoding. The cross-validation analysis of the stimulus-induced dataset (see Materials and Methods "14. Synset pair selection by within-dataset cross-validation") yielded the distribution of accuracies for stimulus-to-stimulus pairwise decoding (Subject 1–3, HVC; shown together with the distribution from label-shuffled decoders). The mean decoding accuracies were 85.6% (95% CI [84.1, 87.0]) for Subject 1, 87.0% (CI [85.4, 88.7]) for Subject 2, and 81.4% (CI [79.4, 83.4]) for Subject 3. The fraction of pairs that showed significantly better decoding performance than chance level (one-tailed binomial test, uncorrected $P < 0.05$) was 94.5% (190/201) for Subject 1, 99.8% (117/118) for Subject 2, and 95.3% (82/86) for Subject 3. The performance was significantly better than that from label-shuffled decoders for all subjects (Wilcoxon rank-sum test, $P < 0.001$ for all three subjects).

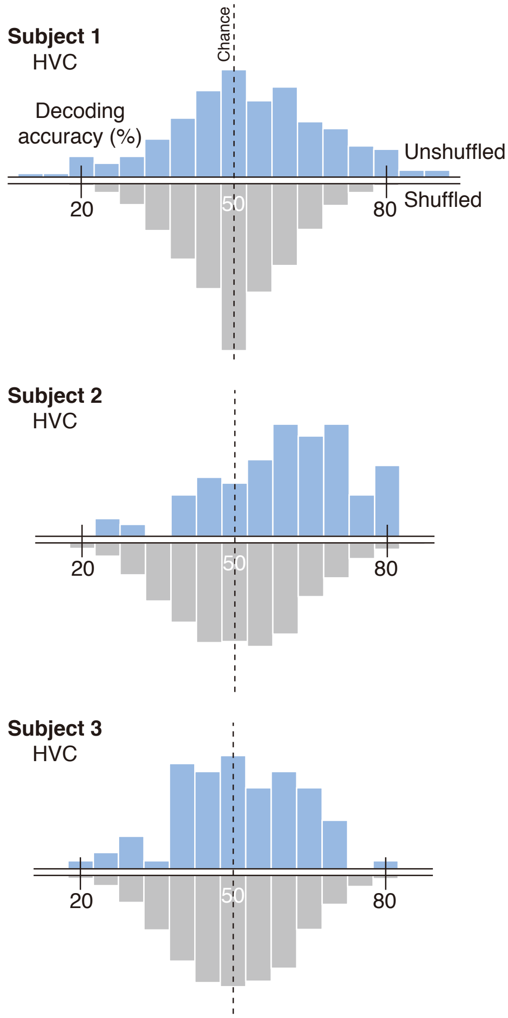

Fig. S6. Sleep-to-sleep pairwise decoding. The cross-validation analysis of the sleep dataset (see Materials and Methods “14. Synset pair selection by within-dataset cross-validation”) yielded the distribution of accuracies for sleep-to-sleep pairwise decoding (Subject 1–3, HVC; shown with the distribution from label-shuffled decoders). The mean decoding accuracies were 52.9% (95% CI [51.2, 54.7]) for Subject 1, 60.1% (CI [58.1, 62.1]) for Subject 2, and 51.3% (CI [49.1, 53.5]) for Subject 3. The fraction of pairs that showed significantly better decoding performance than chance level (one-tailed binomial test, uncorrected $P < 0.05$) was 20.4% (41/201) for Subject 1, 40.7% (48/118) for Subject 2, and 12.8% (11/86) for Subject 3. The results showed that the performance was significantly better than that from label-shuffled decoders for two of three subjects (Wilcoxon rank-sum test, P = 0.002 for Subject 1; $P < 0.001$ for Subject 2; P = 0.302 for Subject 3). Note that although the sleep-to-sleep decoding shows marginally significant accuracies (depending on subjects), it is not as accurate as the stimulus-to-sleep decoding. This is presumably because training samples from the sleep dataset were fewer and noisier than those from the stimulus-induced dataset, and thus decoders were not well trained with the sleep dataset.

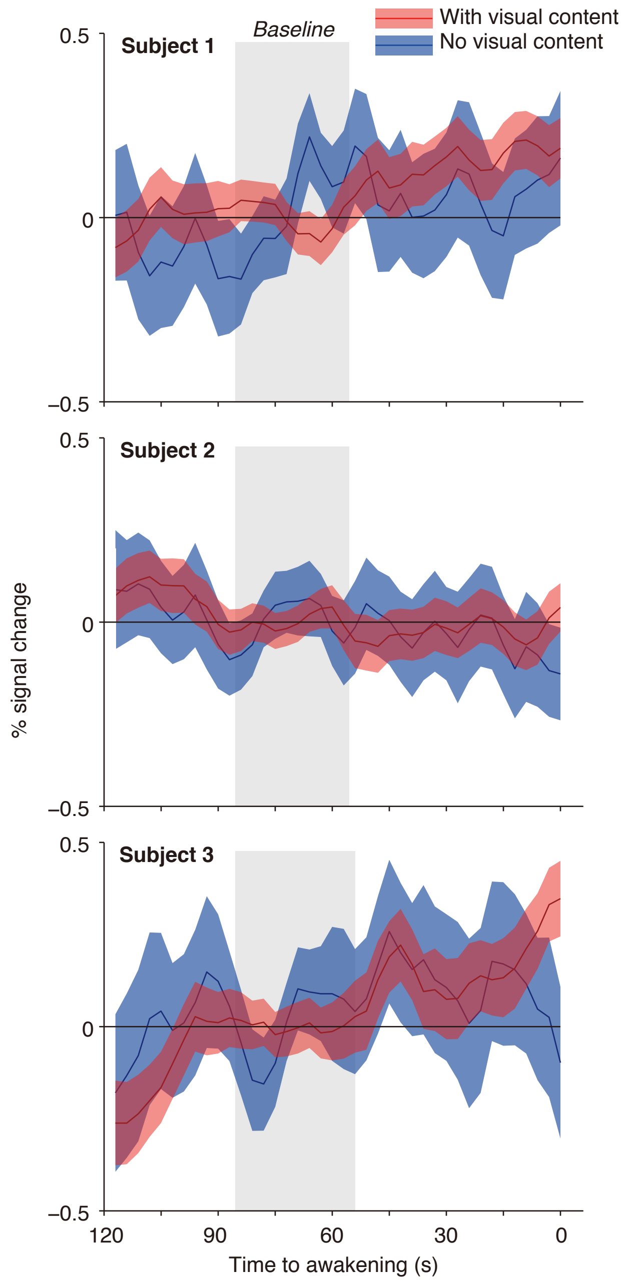

Fig. S7. Time course of BOLD signals. The time course of the averaged BOLD signal in HVC is shown for the awakenings with and without visual content (shades, 95% CI; mean across awakenings). The gray area indicates the period used for the baseline for the normalization. The mean activity level in HVC did not show marked differences toward the timing of awakening whether the subjects reported visual contents or not.

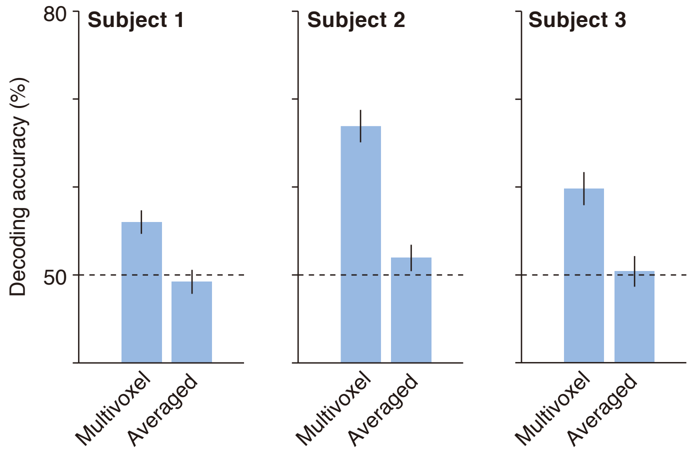

Fig. S8. Decoding with averaged vs. multivoxel activity. The voxel values in each data sample from HVC were averaged (before the z-transformation in each sample), and the averaged activity was used as the input to the pairwise decoders. The decoding accuracy with averaged activity is compared with that from the multivoxel decoders (error bar, 95% CI; averaged across all pairs). The performance with averaged activity is close to the chance level and significantly worse than that with multivoxel activity (Wilcoxon signed-rank test, $P < 0.001$, all subjects).

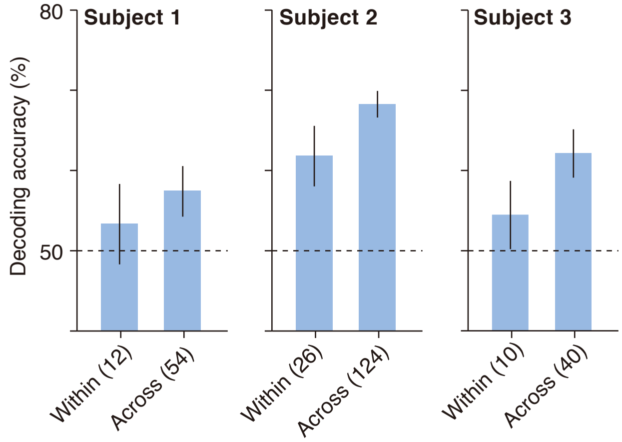

Fig. S9. Decoding within and across meta-categories. Pairwise decoding accuracies grouped into the pairs within and across meta-categories are shown for individual subjects. The numbers of available pairs are denoted in parentheses. Because the *others* category contains a large variety of synsets in terms of semantic similarity, the pairs of synsets in the *others* category were excluded from this analysis. The performance of across-meta-category decoding was better than that of within-meta-category decoding in all subjects, though the significance level varied (Wilcoxon rank-sum test, P = 0.261 for Subject 1; P = 0.003 for Subject 2; P = 0.020 for Subject 3; error bar, 95% CI).

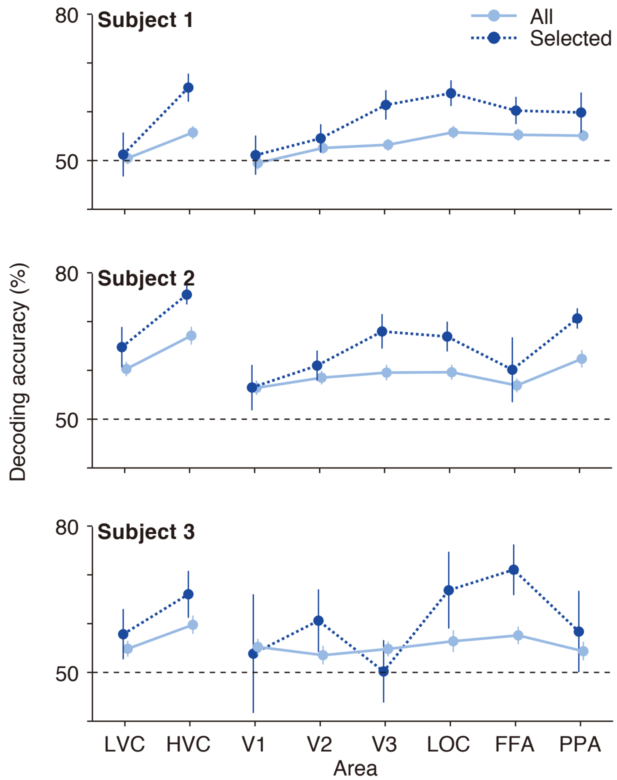

Fig. S10. Pairwise decoding accuracies across visual cortical areas. The decoding accuracies with different visual areas are shown for individual subjects (error bars, 95% CI). Selected pairs were determined on the basis of the cross-validation analysis in each area (the numbers of selected pairs for V1, V2, V3, LOC, FFA, PPA, LVC, and HVC: 24, 22, 29, 38, 36, 22, 27, and 39 pairs for Subject 1; 15, 23, 21, 24, 7, 47, 20, and 47 pairs for Subject 2; 6, 5, 5, 8, 5, 9, 8, and 11 pairs for Subject 3, respectively).

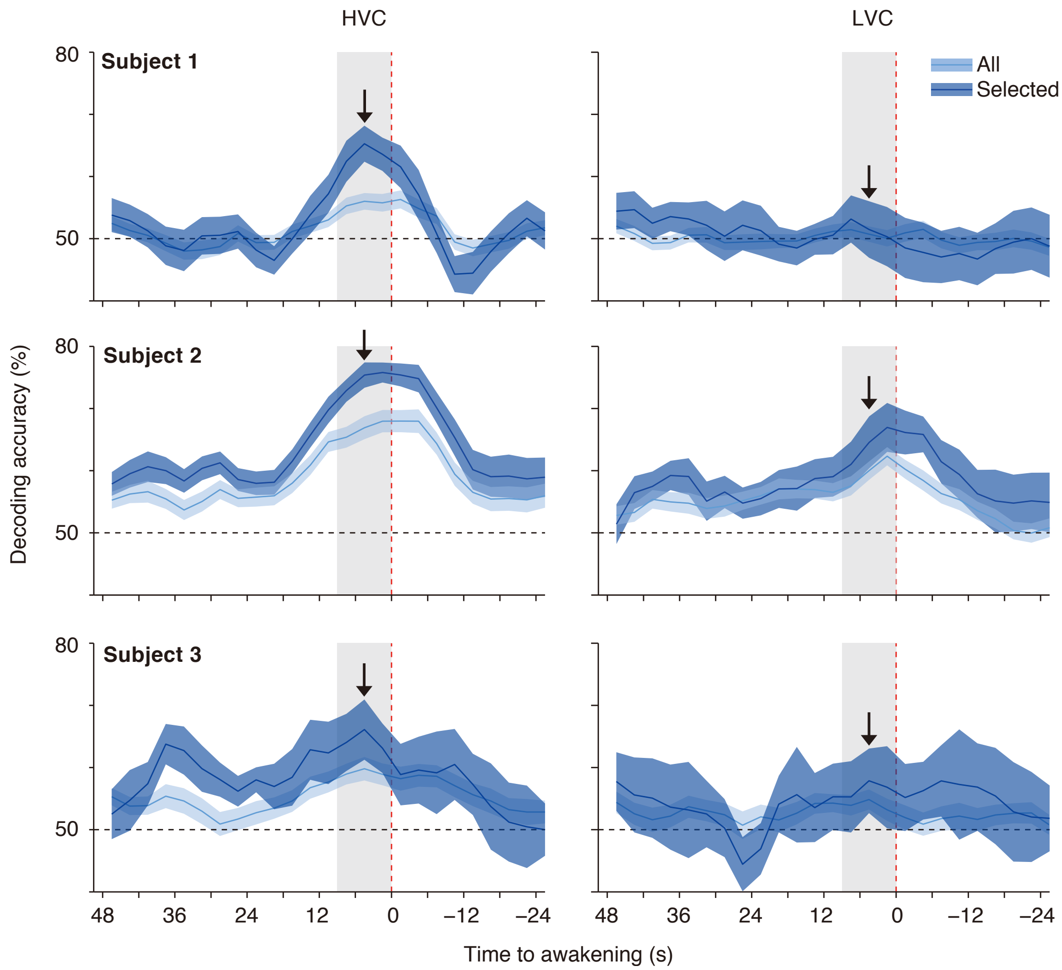

Fig. S11. Time course of pairwise decoding accuracy. The time course of pairwise decoding accuracy is shown for individual subjects (shades, 95% CI; averaged across all or selected pairs). Averages of three fMRI volumes (9 s; HVC or LVC) around each time point were used as inputs to the decoders. The performance is plotted at the center of the window. The gray region indicates the time window used for the main analyses (the arrow denotes the performance obtained from the time window). No corrections for hemodynamic delay were conducted. Note that fMRI signals after awakening may be contaminated with movement artifacts and brain activity associated with mental states during verbal reports. Mental states during verbal reports are unlikely to explain the high accuracy immediately after awakening, because the accuracy profile does not match the mean duration of verbal reports (34 ± 19 s, mean ± SD; three subjects pooled).

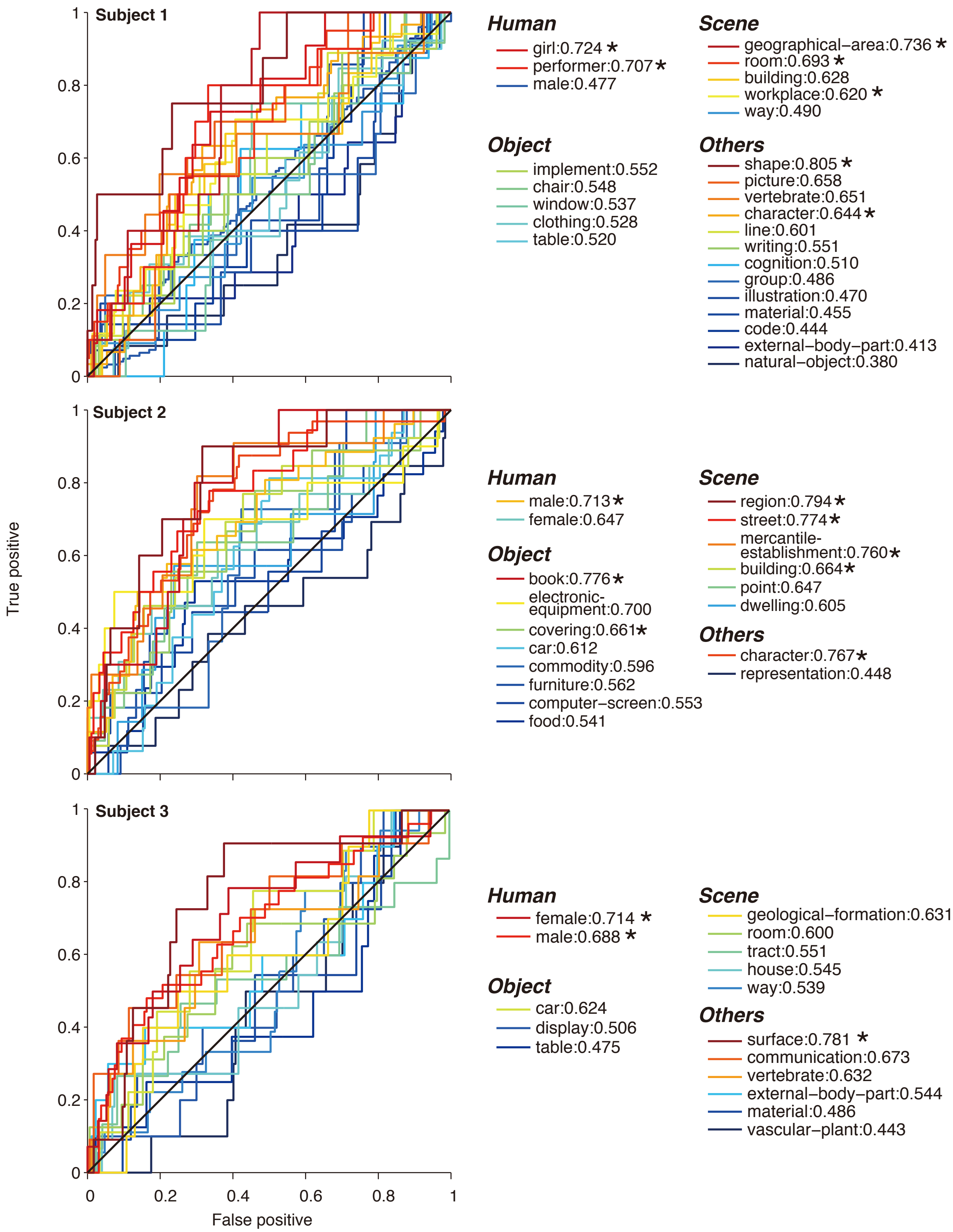

Fig. S12. ROC analysis for all subjects. Above-chance level detection (Wilcoxon rank-sum test, uncorrected *P* < 0.05; denoted by *) was achieved in 7/26 synsets for Subject 1, 8/18 for Subject 2, and 3/16 for Subject 3. Similar results were obtained using the samples with no visual report to represent the absence of each synset (4/26 synsets for Subject 1, 7/18 for Subject 2, and 4/16 for Subject 3).

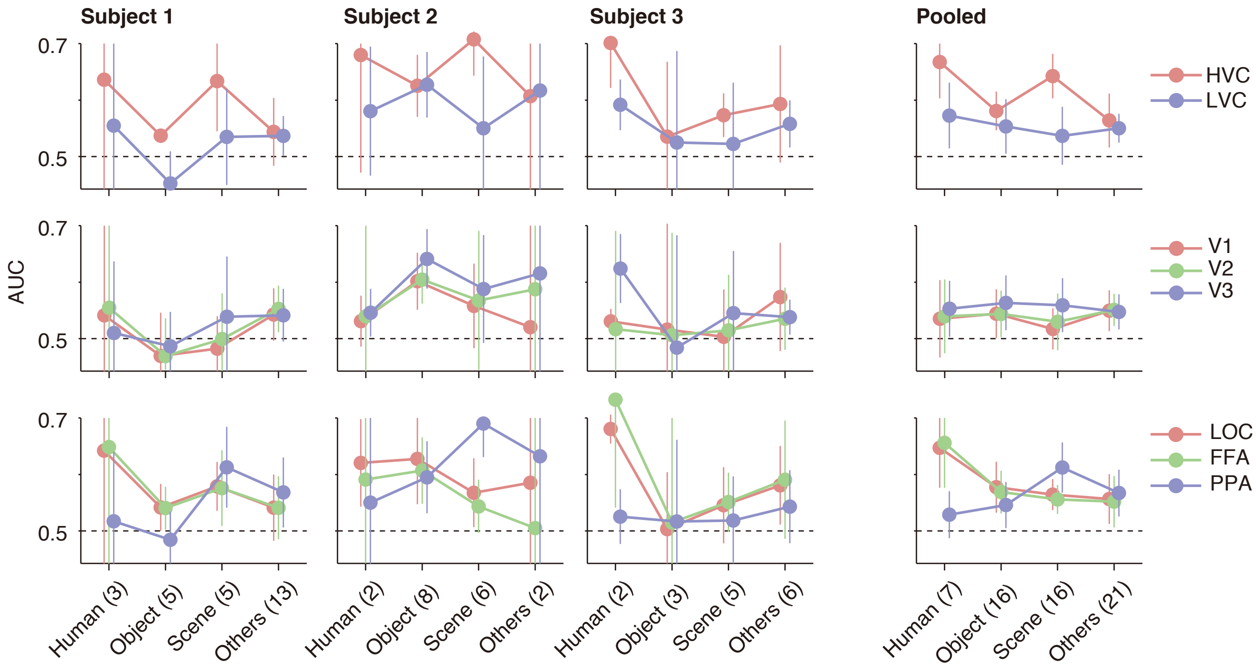

Fig. S13. AUCs for meta-categories in different visual areas. The mean AUC for the synsets within each meta-category is plotted for different visual areas (individual subjects' and pooled results; error bars, 95% CI). The numbers of available synsets are shown in parentheses. Because the sample sizes are small in individual subjects, evaluation with statistical tests is difficult. However, tendencies similar to the pooled results are found in individual subjects: HVC tends to outperform LVC (ANOVA, P = 0.029 for Subject 1, P = 0.159 for Subject 2, and P = 0.139 for Subject 3), and FFA and PPA tend to show better performances with *human* and *scene*, respectively (ANOVA [interaction], P = 0.099 for Subject 1; P = 0.028 for Subject 2; P = 0.044 for Subject 3). Additionally, FFA and LOC showed similar results in individual subjects.

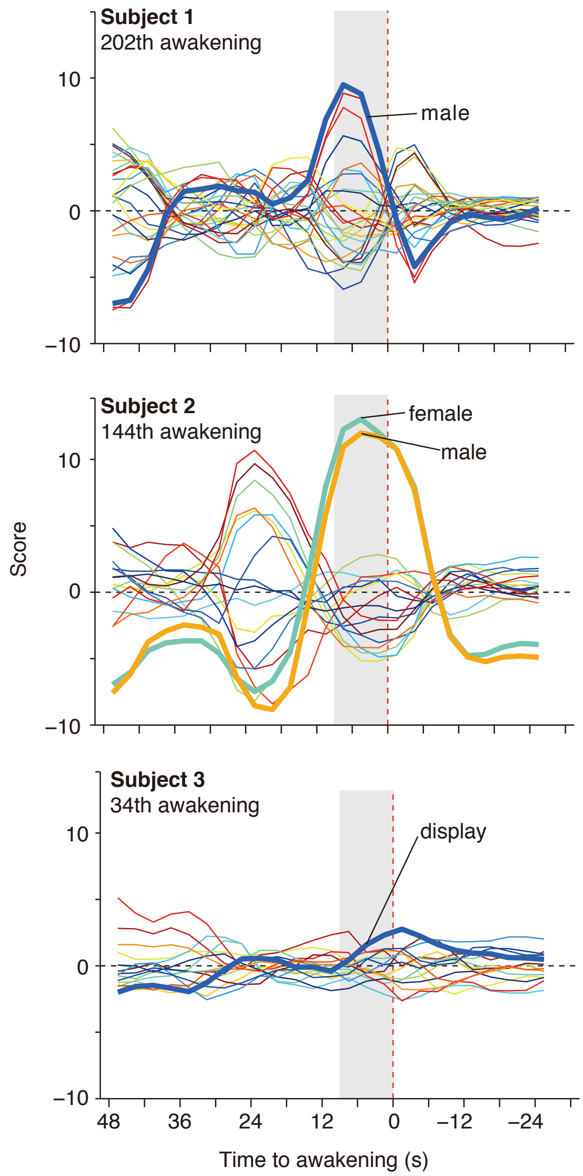

Fig. S14. Examples for the time courses of synset scores. The time courses of synset scores from multilabel decoding analyses are shown for four individual sleep examples. The plots represent the scores for the reported synset(s) (bold line with synset name) and the unreported synsets using the colors in the legend for fig. S12.

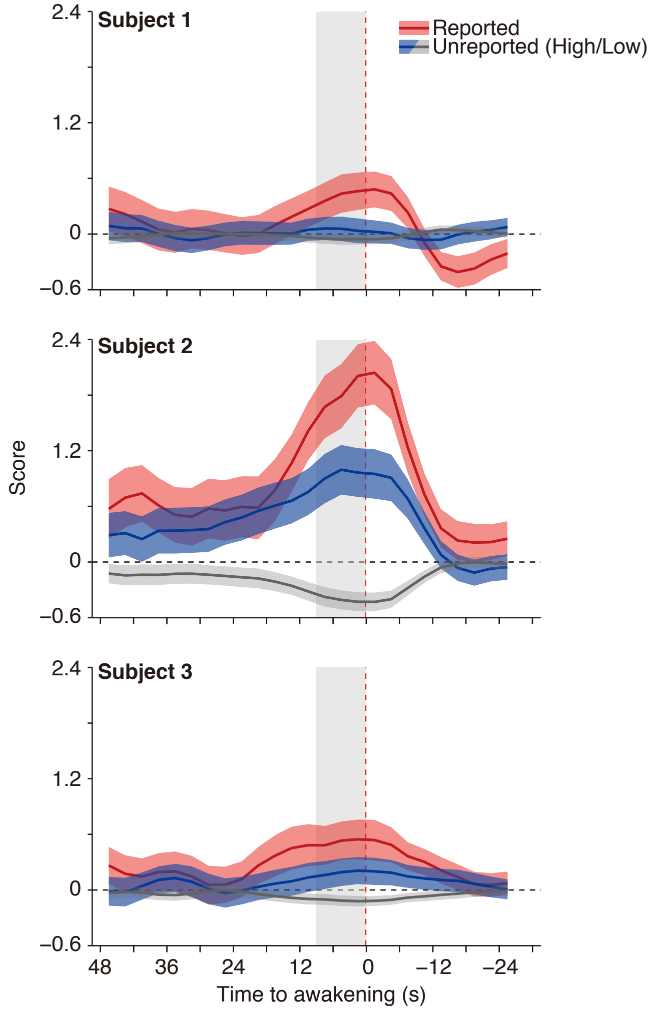

Fig. S15. Time courses of averaged synset scores. Synset scores were averaged across awakenings for reported (red) and unreported synsets with high (blue) and low (gray) co-occurrence in individual subjects (shades, 95% CI).

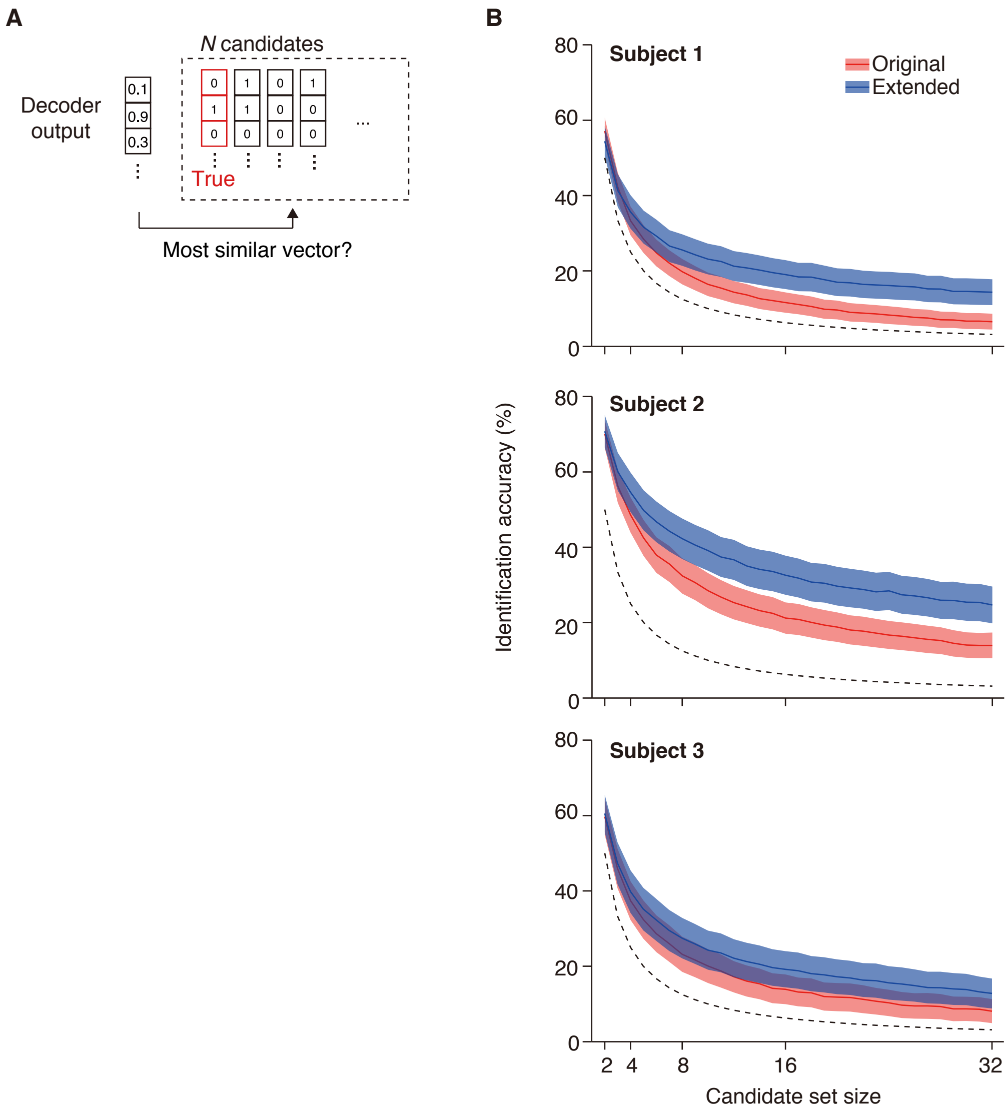

Fig. S16. Identification analysis. (**A**) The correlation coefficients between the score vector from multilabel decoding and each of the candidate vectors consisting of the true visual content vector and a variable number of binary vectors generated randomly with a matched probability were calculated. The vector with the highest correlation coefficient was chosen as the one representing the visual contents. (**B**) Identification performance with original and extended visual content vectors is shown for individual subjects.

Table S1. Examples of Verbal Reports

Subject	Index	Report	Word	Base synset
Subject 1	13	Well, what was that? Two male persons, well, what was that? I cannot remember very well, but there were e-mail texts. There were also characters. E-mail address? Yes, there were a lot of e-mail addresses. And two male persons existed.	male person text character	male (male person) writing (written material, piece of writing) character (grapheme, graphic symbol)
	133	Well, somewhere, in a place like a studio to make a TV program or something, well, a male person ran with short steps, or run, from the left side to the right side. Then, he tumbled. He stumbled over something, and stood up while laughing, and said something. He said something to persons on the left side. So, well, a person ran, and there were a lot of unknown people. I did not saw female persons. There were a group of a lot of people, either male or female. The place was like a studio. Though there are a lot of variety for studios, the studio was a huge room. Since it was a huge room, it was indoor maybe. I saw such a scene in the huge room.	studio room male person people	workplace (work) room male (male person) group (grouping)
Subject 2	54	Yes, I had a dream. Something long. First at some shop. Ah, a bakery shop. I was choosing some merchandise. I took a roll in which a leaf of perilla was put. Then, I went out, and on the street, I saw a person who were doing something like taking a photograph.	shop bakery merchandise roll leaf perilla street person	point mercantile establishment (retail store, sales outlet, outlet) commodity (trade good, good) food (solid food) street
	130	Yes, ah, yes, a female person, well, existed. The person served some foods, umm, like a flight attendant. Then, well, before that scene, I saw a scene in which I ate or saw yogurt, or I saw yogurt or a scence in which yogurt was served. What appeared was the female person and an unknown thing like a refrigerator. Maybe indoor, with colors.	food flight attendant yogurt refrigerator person female person	food (solid food) female (female person) commodity (trade good, good)
Subject 3	114	Well, from the sky, from the sky, well, what was it? I saw something like a bronze statue, a big bronze statue. The bronze statue existed on a small hill. Below the hill, there were houses, streets, and trees in an ordinary way.	sky statue hill house street tree	geological formation (formation) house way vascular plant (tracheophyte)
	186	Well, in the night, somewhere, well, in a restaurant in office building covered with windowpanes, on the big table, someone important looked a menu and chose dishes. There were both male and female persons. Then, there was a night scenery from the window.	restaurant office building windowpane table someone menu male female scenery window	table communication male (male person) female (female person)

Note. Originally, the reports and words were verbally reported in Japanese. They were transcribed into text and were translated into English (verified by a bilingual speaker). Note that not all reported visual words were assigned with a base synset (e.g., *person* in report #133 of Subject 1, and *perilla* in report #54 of Subject 2). This is because 1) the word and its hypernyms did not appear in ten or more reports, or 2) the synset assigned to the word is a hypernym of other synsets (see Materials and Methods "8. Visual content labeling"). On average, 47.7% of reported visual words were assigned with base synsets (49.5%, 50.0%, and 42.0% for Subject 1–3, respectively).

Table S2. List of Base Synsets for Subject 1

Base synset	ID	Definition	Reported word	Count	Meta-category
male (male person)	09624168	a person who belongs to the sex that cannot have babies	gentleman, boy, middle-aged man, old man, young man, male, dandy	127	Human
character (grapheme, graphic symbol)	06818970	a written symbol that is used to represent speech	character, letter	35	Others
room	04105893	an area within a building enclosed by walls and floor and ceiling	booth, conference room, room, toilet	20	Scene
workplace (work)	04602044	a place where work is done	laboratory, recording studio, studio, workplace	17	Scene
external body part	05225090	any body part visible externally	lip, hand, face	17	Others
natural object	00019128	an object occurring naturally; not made by man	leaf, branch, figure, beard, mustache, orange, coconut, moon, sun	13	Others
building (edifice)	02913152	a structure that has a roof and walls and stands more or less permanently in one place	bathhouse, building, house, restaurant, schoolhouse, school	12	Scene
clothing (article of clothing, vesture, wear, wearable, habiliment)	03051540	a covering designed to be worn on a person's body	clothes, baseball cap, clothing, coat, costume, tuxedo, silk hat, hat, T-shirt, kimono, muffler, polo shirt, suit, uniform	13	Object
chair	03001627	a seat for one person, with a support for the back	chair, folding chair, wheelchair	12	Object
picture (image, icon, ikon)	03931044	a visual representation (of an object or scene or person or abstraction) produced on a surface	graphic, picture, image, portrait	12	Others
shape (form)	00027807	the spatial arrangement of something as distinct from its substance	circle, square, quadrangle, box, node, point, dot, tree, tree diagram, hole	11	Others
vertebrate (craniate)	01471682	animals having a bony or cartilaginous skeleton with a segmented spinal column and a large brain enclosed in a skull or cranium	bird, ostrich, raptor, hawk, falcon, eagle, frog, snake, dog, leopard, horse, sheep, monkey, fish, skipjack tuna	11	Others
implement	03563967	instrumentation (a piece of equipment or tool) used to effect an end	trigger, hammer, ice pick, pan, pen, pencil, plunger, pole, pot, return key, stick, wok	10	Object
way	04564698	any artifact consisting of a road or path affording passage from one place to another	hallway, hall, passageway, pedestrian crossing, stairway, street	11	Scene
window	04588739	(computer science) a rectangular part of a computer screen that contains a display different from the rest of the screen	window	11	Object
girl (miss, missy, young lady, young woman, fille)	10129825	a young woman	girl, young woman	11	Human
material (stuff)	14580897	the tangible substance that goes into the makeup of a physical object	water, paper, sand, wood, sheet, leaf, page	11	Others
cognition (knowledge, noesis)	00023271	the psychological result of perception and learning and reasoning	symbol, profile, monster, character	10	Others
group (grouping)	00031264	any number of entities (members) considered as a unit	string, people, pair, pop group, band, calendar, line, forest	10	Others
table	04379243	a piece of furniture having a smooth flat top that is usually supported by one or more vertical legs	desk, stand, table	10	Object
code (computer code)	06355894	(computer science) the symbolic arrangement of data or instructions in a computer program or the set of such instructions	computer code, code	10	Others
writing (written material, piece of writing)	06362953	the work of a writer; anything expressed in letters of the alphabet (especially when considered from the point of view of style and effect)	draft, text, document, written document, clipping, line	10	Others
line	06799897	a mark that is long relative to its width	line	10	Others
illustration	06999233	artwork that helps make something clear or attractive	illustration, figure, Figure	10	Others
geographical area (geographic area, geographical region, geographic region)	08574314	a demarcated area of the Earth	tennis court, campus, playing field, ground, field, lawn, park, parking area, square, public square, town	10	Scene
performer (performing artist)	10415638	an entertainer who performs a dramatic or musical work for an audience	idol, singer, actor, actress, clown, comedian	10	Human

Note. In this study, a representative instance provided by *WordNet* for each synset was used as the name of the base synset. Here, other instances were described in parentheses.

Table S3. List of Base Synsets for Subject 2

Base synset	ID	Definition	Reported word	Count	Meta-category
character (grapheme, graphic symbol)	06818970	a written symbol that is used to represent speech	character	34	Others
male (male person)	09624168	a person who belongs to the sex that cannot have babies	boy, male, male person	27	Human
street	04334599	a thoroughfare (usually including sidewalks) that is lined with buildings	street	21	Scene
car (auto, automobile, machine, motorcar)	02958343	a motor vehicle with four wheels; usually propelled by an internal combustion engine	car, patrol car, police cruiser, used-car	17	Object
food (solid food)	07555863	any solid substance (as opposed to liquid) that is used as a source of nourishment	food, chocolate bar, apple pie, cake, cookie, bread, roll, noodle, tomato, cherry tomato, yogurt, yoghurt	19	Object
building (edifice)	02913152	a structure that has a roof and walls and stands more or less permanently in one place	apartment house, apartment building, building, coffee shop, house, library, school	18	Scene
representation	04076846	a creation that is a visual or tangible rendering of someone or something	map, model, photograph, photo, picture, snowman	14	Others
furniture (piece of furniture, article of furniture)	03405725	furnishings that make a room or other area ready for occupancy	bed, chair, counter, desk, furniture, hospital bed, sofa, couch, table	13	Object
female (female person)	09619168	a person who belongs to the sex that can have babies	girl, wife, female, female person	13	Human
book (volume)	02870092	physical objects consisting of a number of pages bound together	book, notebook	12	Object
point	08620061	the precise location of something; a spatially limited location	bakery, corner, crossing, intersection, crossroad, laboratory, level crossing, studio, bus stop, port, Kobe	12	Scene
commodity (trade good, good)	03076708	articles of commerce	hat, iron, jacket, T-shirt, Kimono, kimono, merchandise, refrigerator, shirt, stove	11	Object
computer screen (computer display)	03085602	a screen used to display the output of a computer to the user	computer screen, computer display	10	Object
electronic equipment	03278248	equipment that involves the controlled conduction of electrons (especially in a gas or vacuum or semiconductor)	amplifier,mobile phone, cell phone, cellular phone, printer, television, TV	11	Object
mercantile establishment (retail store, sales outlet, outlet)	03748162	a place of business for retailing goods	bakery, bookstore, booth, convenience store, department store, shopping center, shopping mall, shop, stall, supermarket	11	Scene
region	08630985	a large indefinite location on the surface of the Earth	garden, downtown, park, parking area, scenery, town, Kobe	10	Scene
covering	03122748	an artifact that covers something else (usually to protect or shelter or conceal it)	accessory, accessories, clothes, covering, flying carpet, hat, jacket, T-shirt, Kimono, kimono, shirt, slipper	10	Object
dwelling (home, domicile, abode, habitation, dwelling house)	03259505	housing that someone is living in	home,house	10	Scene

Table S4. List of Base Synsets for Subject 3

Base synset	ID	Definition	Reported word	Count	Meta-category
male (male person)	09624168	a person who belongs to the sex that cannot have babies	old man, male	28	Human
way	04564698	any artifact consisting of a road or path affording passage from one place to another	entrance, hallway, footpath, penny arcade, pipe, sewer, staircase, stairway, street, tunnel	19	Scene
room	04105893	an area within a building enclosed by walls and floor and ceiling	lobby, hospital room, kitchen, trunk, operating room, room	18	Scene
tract (piece of land, piece of ground, parcel of land, parcel)	08673395	an extended area of land	garden, field, ground, athletic field, green, lawn, grassland, rice paddy, paddy field, park, parking area, savannah, savanna	18	Scene
female (female person)	09619168	a person who belongs to the sex that can have babies	female	14	Human
communication	00033020	something that is communicated by or to or between people or groups	subtitle, text, menu, theatre ticket, poster, character, traffic light, traffic signal, graph, mark	13	Others
vertebrate (craniate)	01471682	animals having a bony or cartilaginous skeleton with a segmented spinal column and a large brain enclosed in a skull or cranium	bird, hawk, eagle, frog, whale, dog, cheetah, horse, water buffalo, sheep, giraffe, fish	12	Others
display (video display)	03211117	an electronic device that represents information in visual form	computer screen, computer display, display, screen, monitor, window	11	Object
surface	04362025	the outer boundary of an artifact or a material layer constituting or resembling such a boundary	ceiling, floor, platform, screen, stage	11	Others
material (stuff)	14580897	the tangible substance that goes into the makeup of a physical object	gravel, cardboard, clay, earth, water, log, paper, playing card	12	Others
car (auto, automobile, machine, motorcar)	02958343	a motor vehicle with four wheels; usually propelled by an internal combustion engine	car, jeep, sport car, sports car, sport car	11	Object
house	03544360	a dwelling that serves as living quarters for one or more families	house	12	Scene
external body part	05225090	any body part visible externally	head, neck, chest, breast, leg, foot, hand, finger, face, human face, face	11	Others
geological formation (formation)	09287968	(geology) the geological features of the earth	bank, beach, gorge, hill, mountain, slope	11	Scene
table	04379243	a piece of furniture having a smooth flat top that is usually supported by one or more vertical legs	counter, desk, operating table, table	10	Object
vascular plant (tracheophyte)	13083586	green plant having a vascular system: ferns, gymnosperms, angiosperms	flower, rice, tree	10	Scene

Table S5. Pairwise Decoding Performance of the Selected Pairs for Subject 1

	Synset pair	Stimulus-to-sleep % correct	Stimulus-to-sleep *P* value	Stimulus-to-stimulus % correct	Sleep-to-sleep % correct	Within/Across
	room vs character	82.0	0.00000	95.3	68.0	Across
	table vs girl	85.7	0.00053	98.4	76.2	Across
	performer vs shape	83.3	0.00113	100.0	88.9	Across
	vertebrate vs natural object	79.0	0.00331	73.4	68.4	Within
	girl vs shape	79.0	0.00331	100.0	89.5	Across
	performer vs window	77.8	0.00545	100.0	72.2	Across
	shape vs room	71.4	0.00851	87.5	78.6	Across
	workplace vs character	66.0	0.01068	98.4	66.0	Across
	picture vs room	70.0	0.01183	85.9	83.3	Across
	performer vs table	75.0	0.01267	96.9	80.0	Across
	illustration vs room	70.4	0.01283	95.3	74.1	Across
	shape vs clothing	73.7	0.01315	79.7	68.4	Across
	girl vs character	65.9	0.01576	96.9	65.9	Across
	shape vs picture	72.2	0.02084	92.2	76.0	Within
	shape vs workplace	68.0	0.02839	95.3	72.0	Across
	performer vs illustration	70.6	0.03278	98.4	76.5	Across
	material vs character	63.2	0.04044	71.9	63.2	Within
	girl vs window	68.4	0.04199	98.4	68.4	Across
	cognition vs shape	68.8	0.04286	95.3	75.0	Within
	writing vs room	65.5	0.04458	90.6	72.4	Across
	chair vs character	62.5	0.04677	84.4	62.5	Across
$P < 0.05$ 22/39 pairs	performer vs writing	68.4	0.04971	98.4	68.4	Across
$P > 0.05$ 17/39 pairs	girl vs room	64.5	0.05300	95.3	80.7	Across
	vertebrate vs chair	66.7	0.05495	82.8	81.0	Across
	performer vs room	63.3	0.07206	96.9	73.3	Across
	girl vs chair	65.2	0.07220	92.2	73.9	Across
	vertebrate vs room	63.0	0.08133	95.3	85.2	Across
	picture vs workplace	63.0	0.08133	81.3	66.7	Across
	girl vs workplace	61.5	0.11966	96.9	69.2	Across
	material vs room	60.0	0.13274	87.5	76.7	Across
	vertebrate vs workplace	57.7	0.20780	95.3	69.2	Across
	window vs room	57.1	0.21319	84.4	78.6	Across
	shape vs male	52.3	0.29521	100.0	57.7	Across
	performer vs code	55.0	0.32736	100.0	80.0	Across
	window vs workplace	48.0	0.58381	79.7	80.0	Across
	performer vs vertebrate	47.4	0.59530	67.2	68.4	Across
	cognition vs vertebrate	47.1	0.60625	62.5	70.6	Within
	external body part vs room	46.9	0.64422	95.3	68.8	Across
	external body part vs workplace	38.7	0.90602	98.4	67.7	Across

Note. Performance of stimulus-to-sleep, stimulus-to-stimulus, and sleep-to-sleep decoding using HVC is shown for the pairs whose stimulus-to-stimulus and sleep-to-sleep decoding accuracies had $P < 0.05$ (uncorrected one-tailed binomial test). The fMRI activity from the 9 s before each awakening was used as input for the stimulus-to-sleep and sleep-to-sleep decoding analyses. The order of the pairs is sorted in the ascending order of P values for stimulus-to-sleep decoding.

Table S6. Pairwise Decoding Performance of the Selected Pairs for Subject 2

	Synset pair	Stimulus-to-sleep % correct	Stimulus-to-sleep P value	Stimulus-to-stimulus % correct	Sleep-to-sleep % correct	Within/Across
	car vs character	91.3	0.00000	92.5	78.3	Across
	building vs character	84.4	0.00000	96.3	80.0	Across
	street vs character	83.3	0.00000	97.5	79.2	Across
	mercantile establishment vs character	86.1	0.00000	95.0	79.1	Across
	male vs character	80.4	0.00000	97.5	67.9	Across
	female vs character	82.2	0.00000	95.0	71.1	Across
	dwelling vs character	82.1	0.00001	96.3	79.5	Across
	book vs car	88.5	0.00002	92.5	76.9	Within
	book vs street	85.7	0.00002	92.5	75.0	Across
	mercantile establishment vs book	94.7	0.00002	88.8	79.0	Across
	point vs character	79.1	0.00004	98.8	69.8	Across
	commodity vs street	85.2	0.00006	93.8	66.7	Across
	book vs building	82.6	0.00018	88.8	69.6	Across
	covering vs street	81.5	0.00023	98.8	66.7	Across
	furniture vs character	75.6	0.00023	98.8	64.4	Across
	car vs male	76.2	0.00026	82.5	78.6	Across
	region vs book	85.0	0.00051	93.8	80.0	Across
	book vs male	75.0	0.00090	95.0	69.4	Across
	covering vs car	80.0	0.00091	91.3	72.0	Within
	covering vs character	73.2	0.00106	98.8	68.3	Across
	commodity vs character	72.1	0.00152	92.5	62.8	Across
	representation vs character	73.0	0.00183	83.8	70.3	Within
	mercantile establishment vs electronic equipment	81.0	0.00184	77.5	76.2	Across
	electronic equipment vs car	76.9	0.00219	80.0	73.1	Within
	electronic equipment vs street	75.0	0.00234	87.5	67.9	Across
	region vs electronic equipment	80.0	0.00298	86.3	75.0	Across
	food vs car	72.7	0.00319	97.5	69.7	Within
	computer screen vs street	74.1	0.00367	86.3	66.7	Across
	street vs male	70.0	0.00399	98.8	77.5	Across
	mercantile establishment vs male	71.4	0.00506	100.0	80.0	Across
	commodity vs car	74.1	0.00542	90.0	77.8	Within
	dwelling vs book	76.5	0.00651	93.8	70.6	Across
	food vs street	69.7	0.00905	96.3	69.7	Across
	representation vs building	69.2	0.01479	66.3	65.4	Across
	region vs computer screen	73.7	0.01707	78.8	68.4	Across
	mercantile establishment vs female	70.8	0.02061	100.0	70.8	Across
	dwelling vs region	70.6	0.03278	67.5	70.6	Within
	representation vs car	66.7	0.03632	90.0	70.4	Across
	food vs male	62.8	0.04137	96.3	67.4	Across
	electronic equipment vs building	66.7	0.04163	77.5	66.7	Across
$P < 0.05$ 42/47 pairs	furniture vs street	64.5	0.04524	71.3	64.5	Across
	female vs street	64.5	0.04524	98.8	67.7	Across
	computer screen vs male	62.9	0.05889	92.5	68.6	Across
	region vs commodity	66.7	0.06332	95.0	81.0	Across
$P > 0.05$ 5/47 pairs	region vs representation	65.2	0.06798	66.3	78.3	Across
	dwelling vs mercantile establishment	61.1	0.15426	66.3	72.2	Within
	electronic equipment vs food	59.3	0.15522	81.3	66.7	Within

Table S7. Pairwise Decoding Performance of the Selected Pairs for Subject 3

		Stimulus-to-sleep		Stimulus-to-stimulus	Sleep-to-sleep	
	Synset pair	% correct	*P* value	% correct	% correct	Within/Across
P < 0.05 9/11 pairs	tract vs male	72.5	0.00142	92.5	62.5	Across
	material vs male	71.1	0.00387	86.3	71.1	Across
	material vs communication	75.0	0.00824	71.3	70.0	Within
	room vs male	69.7	0.00905	95.0	81.8	Across
	communication vs male	67.7	0.01590	92.5	67.7	Across
	geological formation vs communication	71.4	0.01788	83.8	71.4	Across
	female vs room	69.2	0.02092	90.0	69.2	Across
	display vs male	64.9	0.03168	92.5	67.6	Across
	communication vs room	66.7	0.03173	82.5	66.7	Across
P > 0.05 2/11 pairs	geological formation vs room	53.9	0.33934	67.5	65.4	Within
	tract vs room	45.2	0.71926	65.0	64.5	Within

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